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(54) **OPTICAL SYSTEM**

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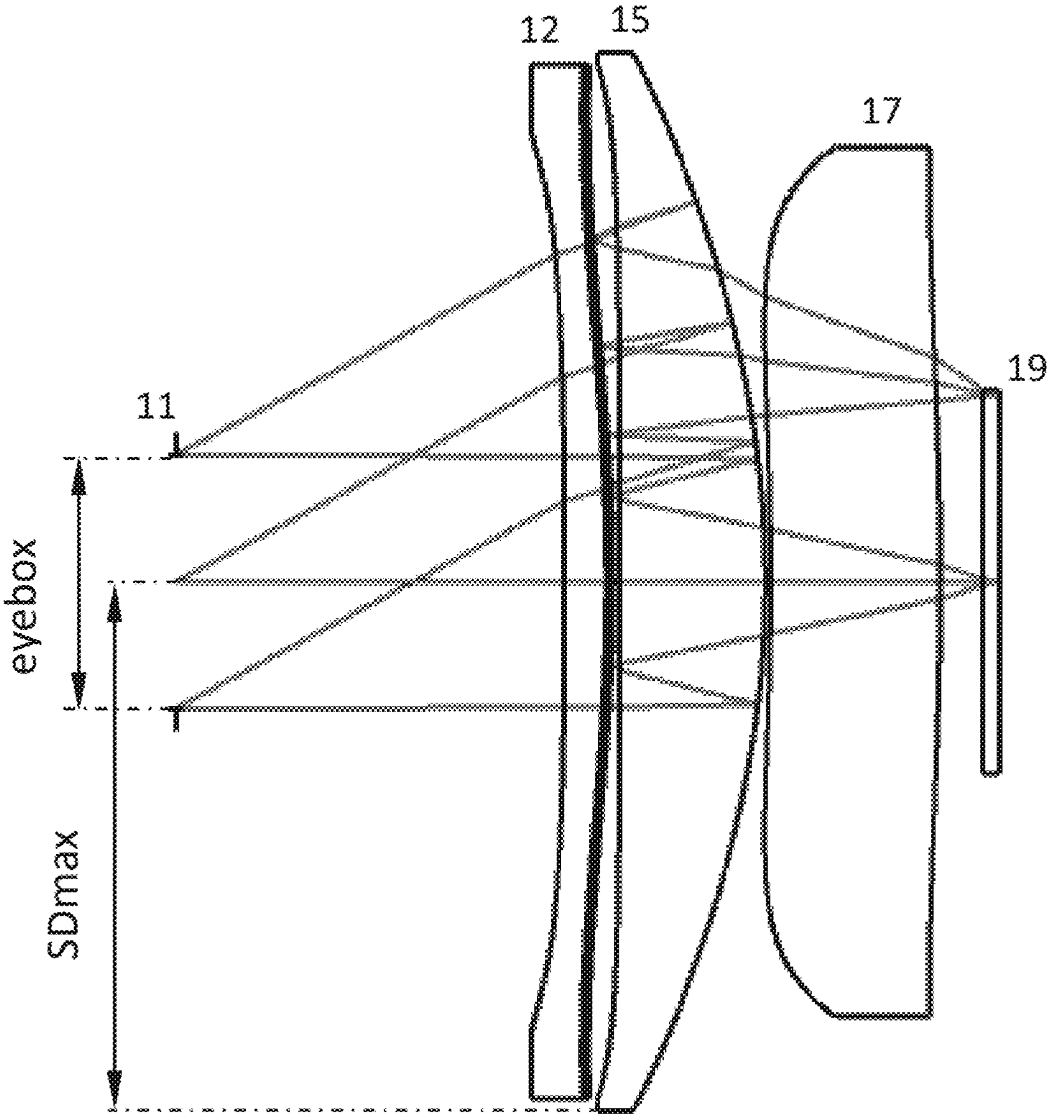
(57) **ABSTRACT**

An optical system is provided, which includes in sequence from a rear side to a front side: an aperture, a first lens, a second lens, a third lens, a circular polarizer, and an image surface. The optical system further satisfies: $5.00 \leq f2/f \leq 9.00$, $1.30 \leq (R1+R2)/(R1-R2) \leq 4.80$, $0.90 \leq R5/R6 \leq 2.80$, and $SDmax \leq 23.00$ mm. f represents a focal length of the optical system, f2 represents a focal length of the second lens, R1 represents a central curvature radius of a rear side surface of the first lens, R2 represents a central curvature radius of the front side surface of the first lens, R5 represents a central curvature radius of a rear side surface of the third lens, R6 represents a central curvature radius of a front side surface of the third lens; and SDmax represents a maximum effective radius of lenses of the optical system.



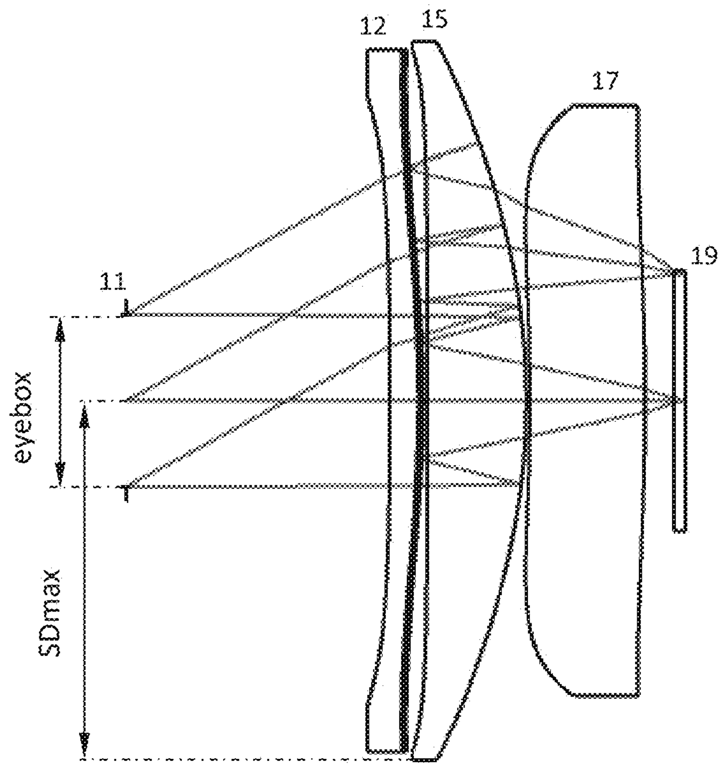

12
15
17
19
11
eyebox
SDmax

FIG. 1

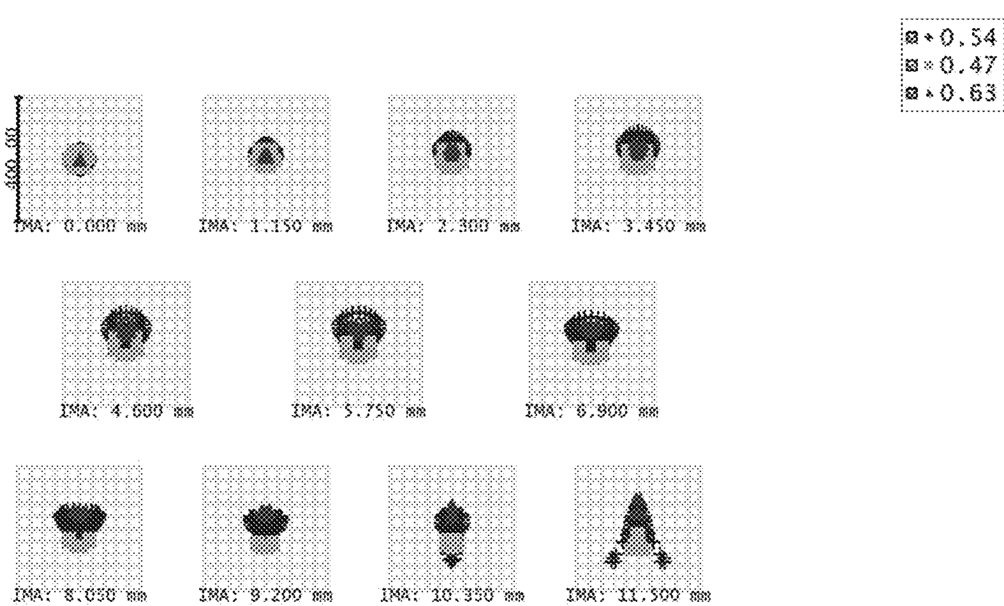

0.54
0.47
0.63
400.00
IMA: 0.000 mm
IMA: 1.150 mm
IMA: 2.300 mm
IMA: 3.450 mm
IMA: 4.600 mm
IMA: 5.750 mm
IMA: 6.900 mm
IMA: 8.050 mm
IMA: 9.200 mm
IMA: 10.350 mm
IMA: 11.500 mm

FIG. 2

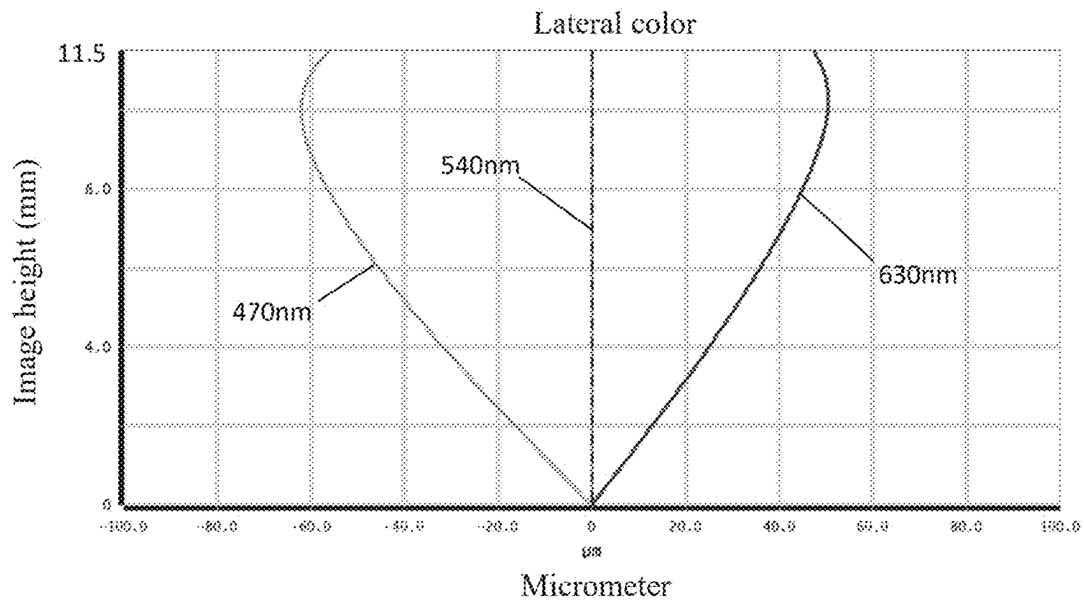

Lateral color
11.5
8.0
4.0
0
Image height (mm)
540nm
470nm
630nm
-100.0
-80.0
-60.0
-40.0
-20.0
0
20.0
40.0
60.0
80.0
100.0
μm
Micrometer

FIG. 3

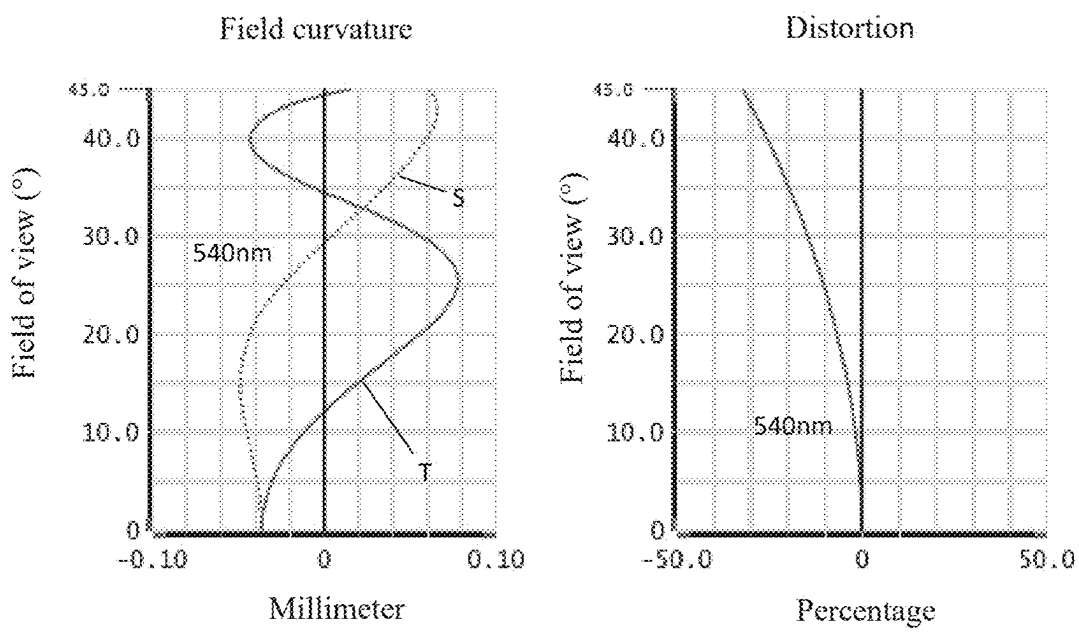

Field curvature
45.0
40.0
30.0
20.0
10.0
0
Field of view (°)
540nm
S
T
-0.10
0
0.10
Millimeter
Distortion
45.0
40.0
30.0
20.0
10.0
0
Field of view (°)
540nm
-50.0
0
50.0
Percentage

FIG. 4

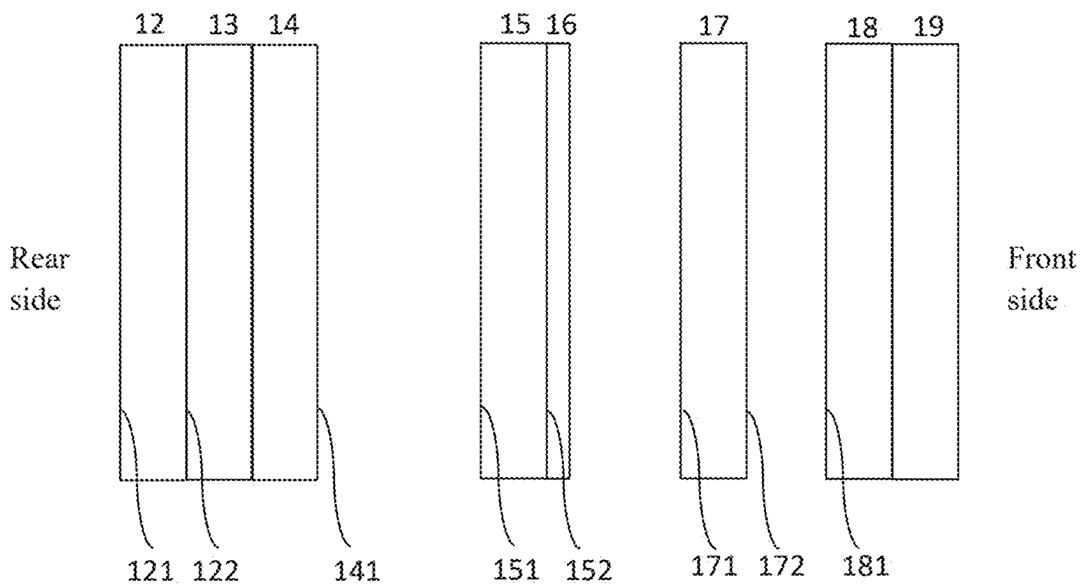

12
13
14
15
16
17
18
19
Rear side
Front side
121
122
141
151
152
171
172
181

FIG. 5

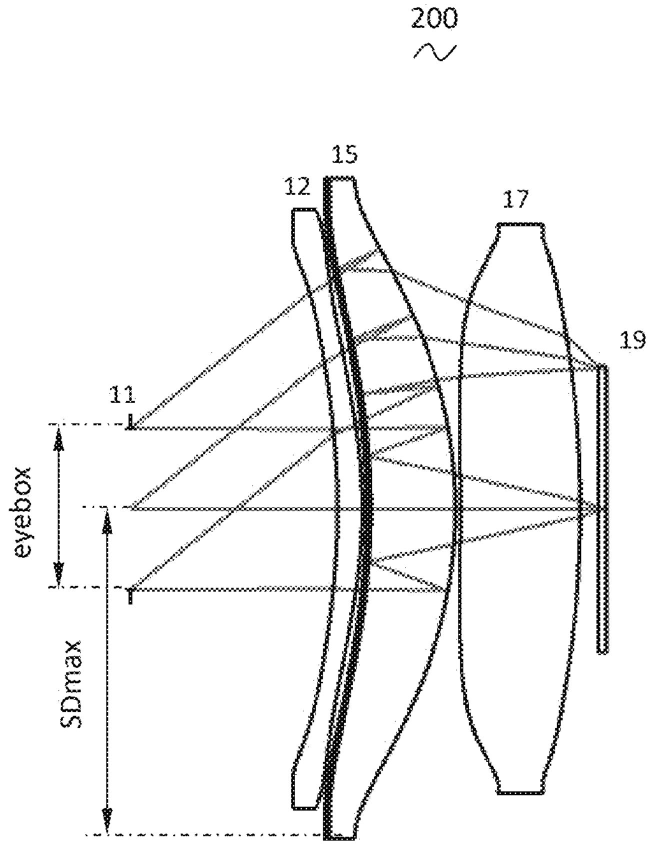

200
15
12
17
19
11
eyebox
SDmax

FIG. 6

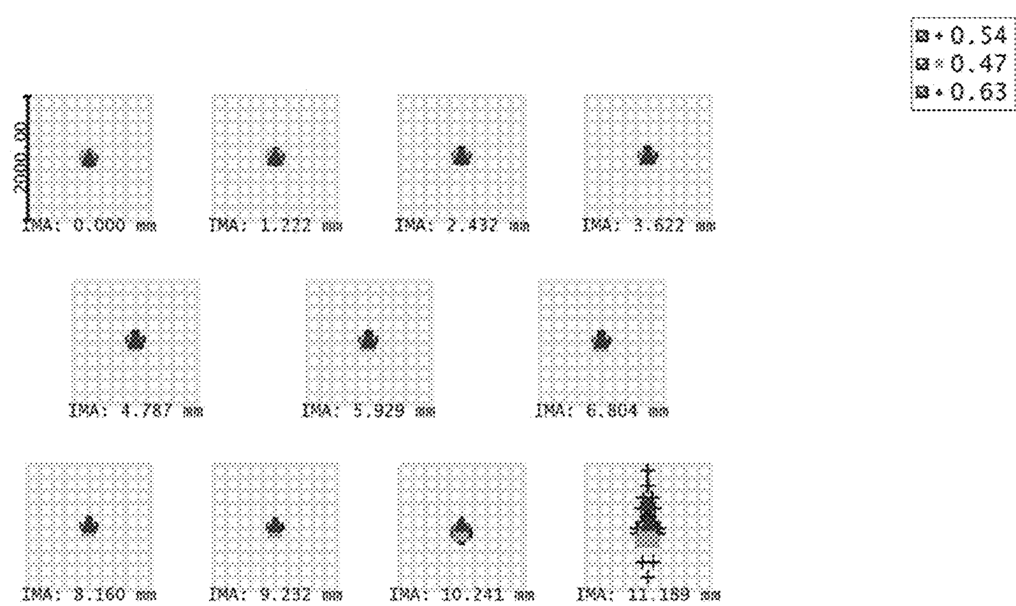

0.54
0.47
0.63
2000.00
IMA: 0.000 mm
IMA: 1.222 mm
IMA: 2.432 mm
IMA: 3.622 mm
IMA: 4.787 mm
IMA: 5.929 mm
IMA: 6.804 mm
IMA: 8.160 mm
IMA: 9.232 mm
IMA: 10.241 mm
IMA: 11.189 mm

FIG. 7

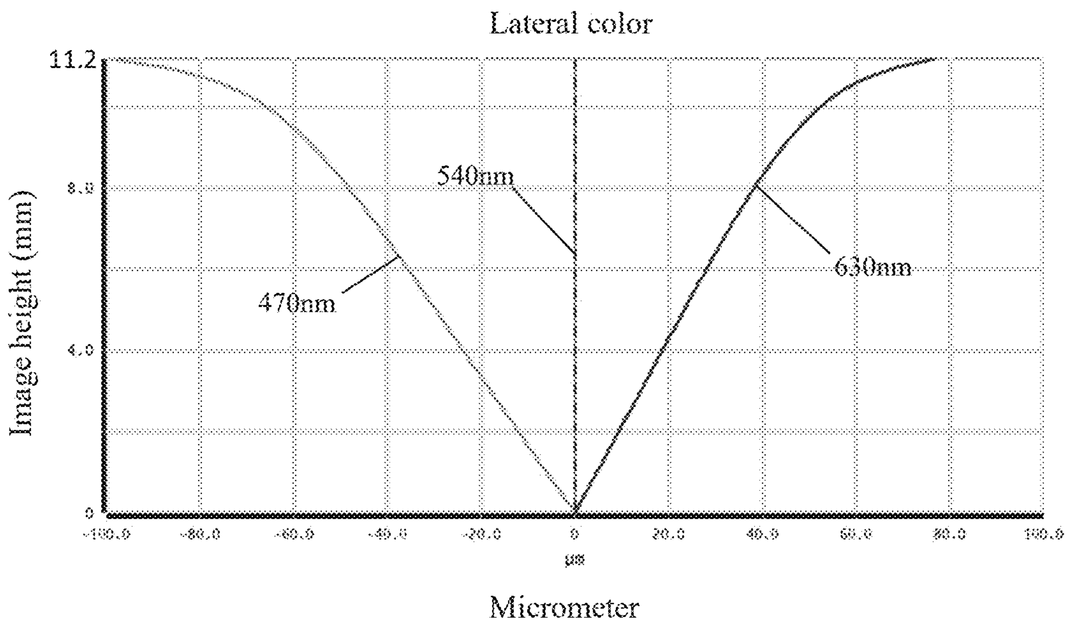

Lateral color
11.2
8.0
4.0
0
Image height (mm)
540nm
470nm
630nm
-100.0
-80.0
-60.0
-40.0
-20.0
0
20.0
40.0
60.0
80.0
100.0
μm
Micrometer

FIG. 8

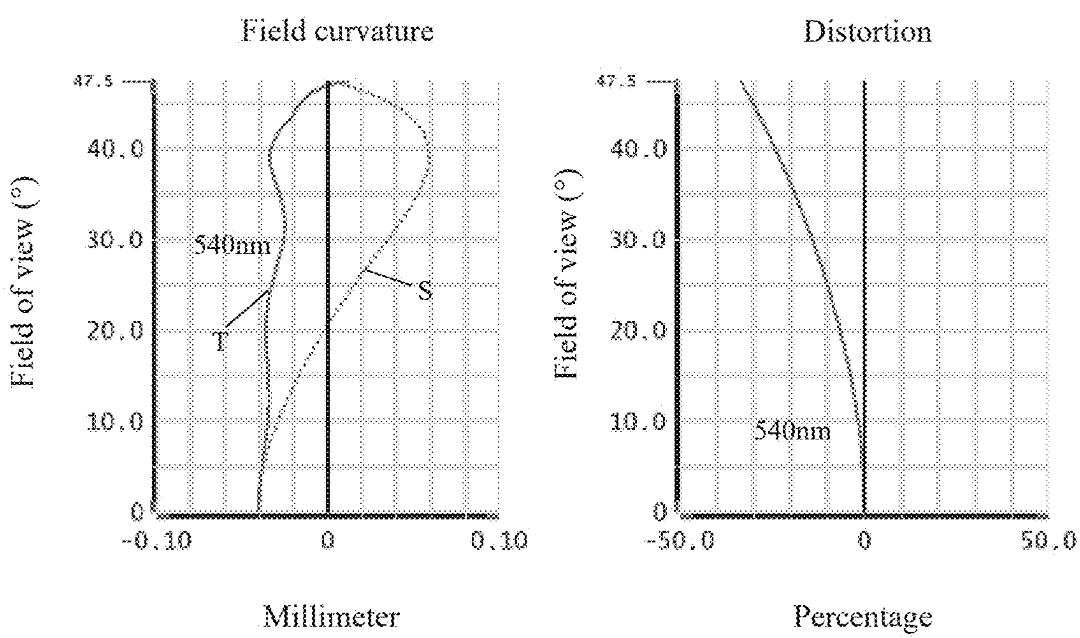

Field curvature
47.5
40.0
30.0
20.0
10.0
0
Field of view (°)
540nm
T
S
-0.10
0
0.10
Millimeter
Distortion
47.5
40.0
30.0
20.0
10.0
0
Field of view (°)
540nm
-50.0
0
50.0
Percentage

FIG. 9

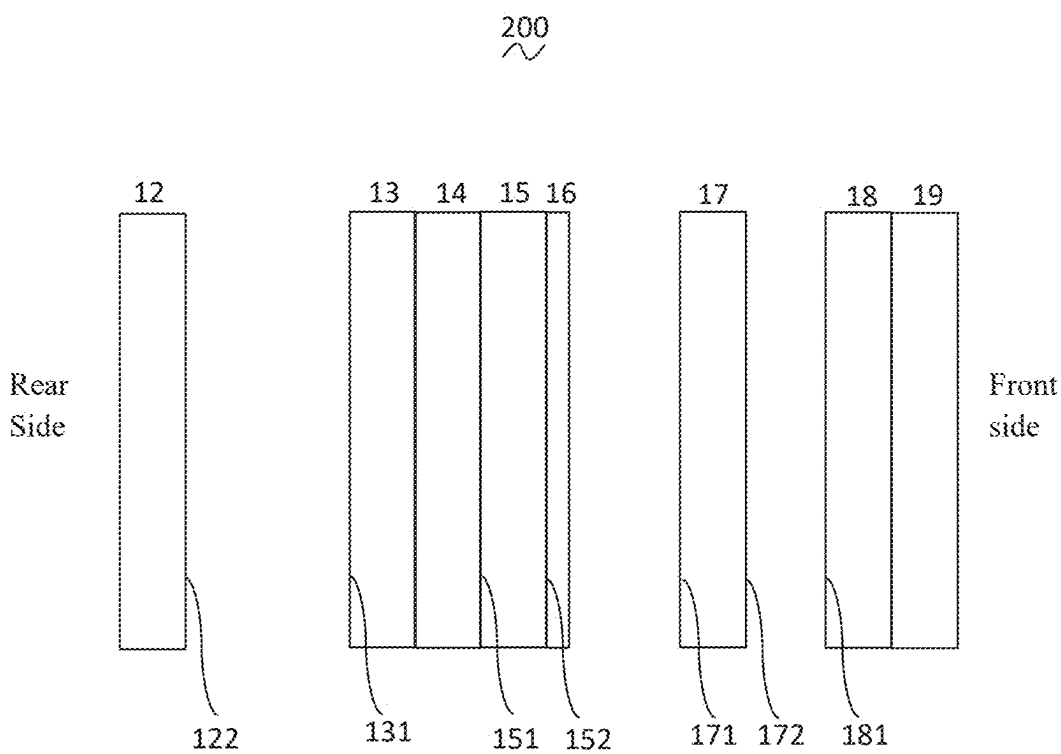

200
12
13
14
15
16
17
18
19
Rear
Side
Front
side
122
131
151
152
171
172
181

FIG. 10

OPTICAL SYSTEM

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of priority under the Paris Convention to Chinese Patent Application No. 202410202541.6, filed on Feb. 23, 2024, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the field of near-eye display technology, and in particular to an optical system.

BACKGROUND

[0003] With the rapid development of technology related to a smart headset device in recent years, the application of an electronic device equipped with an optical lens has become more widespread, the requirements for the optical lens have become more diversified, and the applications in the fields of virtual reality, augmented reality, and mixed reality have grown rapidly. From the perspective of user experience, there is an urgent need for an optical system with both a small size and an excellent imaging method.

SUMMARY

[0004] In view of the above problems, an optical system is provided according to the present disclosure, which has good optical performance and meets the design requirements of miniaturization and lightweight.

[0005] In view of the above problems, an optical system is provided according to the present disclosure, which includes in sequence from a rear side to a front side: an aperture, located on the rear side of the optical system, a first lens, a second lens, a third lens, a circular polarizer, and an image surface. The circular polarizer is attached to a rear side surface of the image surface, a front side surface of the first lens or a rear side surface of the second lens is provided with a composite film including a reflective polarizing coating and a quarter-wave plate, and the reflective polarizing coating is applied on a rear side of the quarter-wave plate. The optical system further satisfies the following conditions: $5.00 \leq f2/f \leq 9.00$, $1.30 \leq (R1+R2)/(R1-R2) \leq 4.80$, $0.90 \leq R5/R6 \leq 2.80$, and $SDmax \leq 23.00$ mm. f represents a focal length of the optical system, f2 represents a focal length of the second lens, R1 represents a central curvature radius of a rear side surface of the first lens, R2 represents a central curvature radius of the front side surface of the first lens, R5 represents a central curvature radius of a rear side surface of the third lens, R6 represents a central curvature radius of a front side surface of the third lens; and SDmax represents a maximum effective radius of lenses of the optical system.

[0006] As an improvement, the optical system further satisfies the following condition: $eyebox \geq 12$ mm, where eyebox represents a size of an eye box of the optical system.

[0007] As an improvement, the optical system further satisfies the following condition: $eyerelief \leq 16.50$ mm, where eyerelief represents an on-axis distance from a human eye to the rear side surface of the first lens.

[0008] As an improvement, the optical system further satisfies the following condition: $TL \leq 18.40$ mm, where TL represents an on-axis distance from the rear side surface of the first lens to the image surface.

[0009] As an improvement, the optical system further satisfies the following condition: $TTL \leq 34.80$ mm, where TTL represents a total track length TTL of the optical system.

[0010] As an improvement, the rear side surface and the front side surface of the first lens, the rear side surface and a front side surface of the second lens, and the rear side surface and the front side surface of the third lens are aspherical.

[0011] As an improvement, the optical system further satisfies the following condition: $85.00° \leq FOV \leq 95.00°$, where FOV represents a field of view of the optical system.

[0012] As an improvement, a front side surface of the second lens is coated with a semi-transparent and semi-reflective film.

[0013] As an improvement, each of a transmittance and a reflectivity of the semi-transparent and semi-reflective film ranges from 40% to 60%.

[0014] As an improvement, a transmittance of the reflective polarizing coating is greater than or equal to 95%.

[0015] As an improvement, the optical system further satisfies the following condition: $MIST \leq 35.00\%$, where MIST represents a distortion of the optical system.

[0016] As an improvement, the optical system further satisfies the following condition: $LC \leq 100$ μm, where LC represents a lateral color of the optical system.

[0017] As an improvement, the optical system further satisfies the following condition: $TTL/f \leq 2.10$, where TTL represents a total track length of the optical system.

[0018] As an improvement, the image surface is a display with a size ranging from 1.0 inch to 1.3 inches.

[0019] The present disclosure has the following beneficial effects: the composite film including the reflective polarizing coating and the quarter-wave plate is arranged on the front side surface of the first lens or the rear side surface of the second lens, and the semi-transparent and semi-reflective film is arranged on the front side surface of the second lens, so that an optical path folding structure is realized through the participation of two lenses, and the total track length TTL of the optical system is greatly reduced, thereby reducing a volume of an optical imaging module and meeting the design requirements of miniaturization and lightweight of a VR device. In addition, the optical system according to the present disclosure enables a user to obtain the best display effect without complicated adjustment, with both small volume and high imaging performance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] In order for a clearer illustration of technical solutions in embodiments of the present disclosure or the conventional technology, drawings used in the description of the embodiments or the conventional technology are described briefly hereinafter. Apparently, the drawings described in the following illustrate some embodiments of the present disclosure, other drawings may be obtained by those ordinarily skilled in the art based on these drawings without any creative effort.

[0021] FIG. **1** is a schematic structural diagram of an optical system according to a first embodiment of the present disclosure;

[0022] FIG. **2** is a schematic diagram of a dot column of the optical system shown in FIG. **1**;

[0023] FIG. **3** is a schematic diagram of a lateral color of the optical system shown in FIG. **1**;

[0024] FIG. **4** is a schematic diagram of a field curvature and a distortion of the optical system shown in FIG. **1**;

[0025] FIG. **5** is a schematic diagram of the optical system including a film structure shown in FIG. **1**;

[0026] FIG. **6** is a partially schematic structural diagram of an optical system according to a second embodiment of the present disclosure;

[0027] FIG. **7** is a schematic diagram of a dot column of a camera optical lens shown in FIG. **6**.

[0028] FIG. **8** is a schematic diagram of a lateral color of the camera optical lens shown in FIG. **6**;

[0029] FIG. **9** is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. **6**; and

[0030] FIG. **10** is a schematic diagram of the optical system including a film structure shown in FIG. **6**.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0031] In order to make the purpose, technical solutions and advantages of the present disclosure clear, the embodiments of the present disclosure are described in detail in conjunction with the drawings. However, it can be understood by those skilled in the art that in the embodiments of the present disclosure, many technical details are proposed to enable the reader to better understand the present disclosure. However, even without these technical details and various changes and modifications based on the following embodiments, the technical solutions claimed by the present disclosure can be realized.

[0032] The technical solutions in the embodiments of the present disclosure are described clearly and completely in conjunction with the drawings of the embodiments of the disclosure hereinafter. It is apparent that the described embodiments are only some rather than all embodiments of the present disclosure. Any other embodiments obtained by those skilled in the art based on the embodiments in the present disclosure without any creative effort shall fall within the protection scope of the present disclosure.

[0033] Referring to FIG. **1** to FIG. **10**, an optical system **100** shown in FIG. **1** and an optical system **200** shown in FIG. **6** are provided according to the technical solutions of the present disclosure. The optical systems **100**, **200** each include three lenses. Specifically, the optical systems **100**, **200** each include in sequence from a rear side to a front side: an aperture **11**, a first lens **12**, a reflective polarizing coating **13**, a quarter-wave plate **14**, a second lens **15**, a semi-transparent and semi-reflective film **16**, a third lens **17**, a circular polarizer **18**, and an image surface **19**. According to the present disclosure, the image surface **19** is a display with a size ranging from 1.0 inch to 1.3 inches. It should be noted that according to the present disclosure, the rear side is a human eye side and the front side is a display side, the front side surface refers to a surface facing the display side in an optical axis direction, and the rear side refers to a surface facing the human eye side in the optical axis direction.

[0034] Specifically, the light emitted from the image surface **19** passes through the circular polarization **18** to form left-handed circularly polarized light LCP incident on the third lens **17**. A front side surface of the second lens **15** is coated with the semi-transparent and semi-reflective film **16**. The left-handed circularly polarized light LCP emitted from a rear side surface of the third lens **17** is incident on the semi-transparent and semi-reflective film **16**, and part of the light is reflected and part of the light is incident on the second lens **15**. In this case, the light incident on the second lens **15** is the left-handed circularly polarized light LCP.

[0035] When a front side surface of the first lens **12** is provided with a composite film including the reflective polarizing coating **13** and the quarter-wave plate **14**, the reflective polarizing coating **13** is closer to the front side surface of the first lens **12** than the quarter-wave plate **14**. Left-handed circularly polarized light LCP emitted from the second lens **15** passes through the quarter-wave plate **14** for the first time and is converted into linearly polarized S-light. The linearly polarized S-light is subsequently reflected at the reflective polarizing coating **13**. In this case, the reflected light is still the linearly polarized S-light. After passing through the quarter-wave plate **14** for the second time, the linearly polarized S-light is converted into the left-handed circularly polarized light LCP incident on the second lens **15** for the second time, part of the left-handed circularly polarized light LCP is reflected at the semi-transparent and semi-reflective film **16**, and the reflected light is converted into right-handed circularly polarized light RCP incident on the second lens **15** for the third time. The right-handed circularly polarized light RCP emitted from the second lens **15** is incident on the quarter-wave plate **14**, is converted into the linearly polarized P-light after passing through the quarter-wave plate **14**, and is incident on the reflective polarizing coating **13**. Since the reflective polarizing coating **13** has the characteristics of reflecting the linearly polarized S-light and transmitting the linearly polarized P-light, the linearly polarized P-light is transmitted to the first lens **12** and is incident on the aperture **11**. A position of the aperture **11** is a position simulating a surface of a human eye.

[0036] When a rear side surface of the second lens **15** is provided with a composite film including the reflective polarizing coating **13** and the quarter-wave plate **14**, the reflective polarizing coating **13** is further away from the rear side surface of the second lens **15** than the quarter-wave plate **14**, Left-handed circularly polarized light LCP emitted from the second lens **15** passes through the quarter-wave plate **14** for the first time and is converted into linearly polarized S-light, and is subsequently reflected at the reflective polarizing coating **13**. In this case, the reflected light is still the linearly polarized S-light. After passing through the quarter-wave plate **14** for the second time, the linearly polarized S-light is converted into the left-handed circularly polarized light LCP incident on the second lens **15** for the second time, part of the left-handed circularly polarized light LCP is reflected at the semi-transparent and semi-reflective film **16**, and the reflected light is converted into right-handed circularly polarized light RCP incident on the second lens **15** for the third time. The right-handed circularly polarized light RCP emitted from the second lens **15** is incident on the quarter-wave plate **14**, is converted into the linearly polarized P-light after passing through the quarter-wave plate **14**, and is incident on the reflective polarizing coating **13**. Since the reflective polarizing coating **13** has the characteristics of reflecting the linearly polarized S-light and transmitting the linearly polarized P-light, the linearly polarized P-light is transmitted to the first lens **12** and is incident on the aperture **11**.

[0037] The front side surface of the first lens **12** is provided with the composite film including the reflective polarizing coating **13** and the quarter-wave plate **14**, and an optical path folding structure is used, which reduces the volume of the optical system. The semi-transparent and semi-reflective film **16** is arranged on the front side surface of the second lens **15**, and the first lens **12** and the second lens **15** are used to participate in the folding scheme of the optical path scheme, which greatly reduces the total track length of the optical system and is beneficial to the development trend of miniaturization and lightweight of a VR device.

[0038] A focal length of the optical system is defined as f, and a focal length of the second lens **15** is defined as f2. The optical system further satisfies the following condition: $5.00 \le f2/f \le 9.00$. A ratio of the focal length of the second lens **15** to the total focal length of the optical system is specified. Within the range of the condition, the off-axis aberration of the field of view can be corrected by the distribution of the focal length.

[0039] A central curvature radius of the rear side surface of the first lens **12** is defined as R1, and a central curvature radius of the front side surface of the first lens **12** is defined as R2. The optical system further satisfies the following condition: $1.30 \le (R1+R2)/(R1-R2) \le 4.80$. A shape of the first lens is specified. Within the range of the condition, it is beneficial to correcting the aberration of light during reflection by reasonably controlling the surface type, and is beneficial to shortening the total track length of the optical system.

[0040] A central curvature radius of the rear side surface of the third lens **17** is defined as R5, and a central curvature radius of the front side surface of the third lens **17** is defined as R6. The optical system further satisfies the following condition: $0.90 \le R5/R6 \le 2.80$. Within the range of the condition, an angle of light emitting from the third lens **17** can be effectively increased by controlling the surface type of the third lens **17** on a screen side, so that the optical system has a large field of view, thereby providing a large eye movement range.

[0041] A maximum effective radius of lenses in optical system is defined as SDmax. The optical system further satisfies the following condition: $SDmax \le 23.00$ mm, thereby being beneficial to reducing a volume of the optical system within the range of the condition.

[0042] A size of an eye box of the optical system is defined as eyebox. The optical system further satisfies the following condition: $eyebox \ge 12$ mm. Within the range of the condition, a user can see the best display effect at the best position without complicated adjustment, the FOV is increased, and the FOV can reach 90°. For further understanding, the size eyebox of the eye box is referred to as a sum of an entrance pupil diameter $ENPD \pm 4$ mm and an eye movement range $eyeshift \pm 4$ mm of the optical system. By setting the condition that $eyebox \ge 12$ mm, the performance of the full field of view of the optical system is better, and the display experience of the users caused by poor adjustment of the glass position is improved.

[0043] An on-axis distance from the human eye to the rear side surface of the first lens **12** is defined as eyerelief, that is, an on-axis distance from the aperture **11** to the rear side surface of the first lens **12** is defined as eyerelief, which is a space where other structure (such as a mechanical mechanism, a glass, etc.) can be placed. The optical system further satisfies the following condition: $eyerelief \le 16.50$ mm. Within the range of the condition, the total track length of the optical system is smaller under the condition of meeting the eye smoothness, which is beneficial to miniaturization.

[0044] An on-axis distance from the rear side surface of the first lens **12** to the image surface **19** is defined as TL. The optical system further satisfies the following condition: $TL \le 18.40$ mm, thereby being beneficial to the miniaturization of the optical system within the range of the condition.

[0045] A total track length of the optical system is defined as TTL, which is an on-axis distance from the human eye to the image surface **19**, that is, an on-axis distance from the aperture **11** to the image surface **19**. The optical system further satisfies the following condition: $TTL \le 34.80$ mm, thereby being beneficial to the miniaturization of the optical system within the range of the condition.

[0046] The rear side surface and the front side surface of the first lens **12**, the rear side surface and the front side surface of the second lens **15**, and the rear side surface and the front side surface of the third lens **17** are aspherical. By adjusting the focus position of the displayed image with the design of aspheric surface, the lateral color and the distortion of the displayed image are reduced and the imaging quality is improved.

[0047] According to the present disclosure, each of a transmittance and a reflectivity of the semi-transparent and semi-reflective film **16** ranges from 40.00% to 60.00%. For example, in various embodiments, a ratio from the transmittance to the reflectivity may be 50:50, 40:60, 60:40, etc.

[0048] According to the present disclosure, a transmittance of the reflective polarizing coating **13** is greater than or equal to 95%, and the high transmittance improves the light efficiency of the optical system and increases the display brightness.

[0049] A distortion of the optical system is defined as MIST. The optical system further satisfies the following condition: $MIST \le 35.00\%$. Within the range of the condition, the distortion of the optical system is relatively small, which provides the user with a more realistic VR device.

[0050] A lateral color of the optical system is defined as LC. The optical system further satisfies the following condition: $LC \le 100$ μm. Within the range of the condition, the lateral color of the optical system is small, which provides the user with a more realistic device.

[0051] A focal length of the optical system is defined as f. The optical system further satisfies the following condition: $TTL/f \le 2.10$, thereby being beneficial to reducing a volume of the optical system within the range of the condition.

First Embodiment

[0052] The optical system **100** according to the present disclosure will be described with an example. The symbols recorded in each example are as follows. The units of the focal length, the on-axis distance, the central curvature radius, the on-axis thickness, the inflexion points and the arrest point are mm.

[0053] The optical system **100** according to a first embodiment of the present disclosure is shown in FIG. **1** to FIG. **5**.

[0054] As shown in FIG. **5**, in the first embodiment, the composite film including the reflective polarizing coating **13** and the quarter-wave plate **14** is arranged on the front side surface **122** of the first lens **12**.

[0055] Tables 1 and 2 show the design data of the optical system **100** according to the first embodiment of the present disclosure.

TABLE 1

	R		d		nd		vd
OBJECT	∞		−1437.5				
Aperture	∞	d0	16.400				
R1	−528.090	d1	1.748	n1	1.658	v1	21.00
R2	−83.460	d2	0.166		1.492		57.44
R2	−83.460	d3	0.122		1.492		57.44
R2	−83.460	d4	0.289				
R3	−1395.925	d5	6.098	n2	1.544	v2	56.11
R4	−46.238	d6	−6.098				
R3	−1395.925	d7	−0.289				
R2	−83.460	d8	−0.122		1.492		57.44
R2	−83.460	d3	0.122				
R2	−83.460	d4	0.289				
R3	−1395.925	d5	6.098	n2	1.544	v2	56.11
R4	−46.238	d9	0.248				
R5	−141.445	d10	7.155	n3	1.544	v3	56.11
R6	−151.139	d11	1.811				
Circular polarizer		d12	0.671	ng	1.517	vg	64.17
Image surface							

[0056] The meanings of various symbols are described as follows:

[0057] R: curvature radius at a center of an optical surface;

[0058] R1: central curvature radius of the rear side surface of the first lens **18**;

[0059] R2: central curvature radius of the front side surface of the first lens **18**;

[0060] R3: central curvature radius of the rear side surface of the second lens **15**;

[0061] R4: central curvature radius of the front side surface of the second lens **15**;

[0062] R5: central curvature radius of the rear side surface of the third lens **14**;

[0063] R6: central curvature radius of the front side surface of the third lens **14**;

[0064] d: on-axis thickness of the lens, and an on-axis distance between the lenses (in order to understand the optical path easily, the value with the light propagating from the rear side to the front side is set as a positive value, and the value with the light from the front side to the rear side is set as a negative value);

[0065] d0: The on-axis distance from the aperture **11** to the rear side surface **121** of the first lens **12**;

[0066] d1: on-axis thickness of the first lens **12**;

[0067] d2: on-axis thickness of the reflective polarizing coating **13**;

[0068] d3: on-axis thickness of the quarter-wave plate **14**;

[0069] d4: on-axis distance from the front side surface **141** of the quarter wave film **14** to the rear side surface **151** of the second lens **15**;

[0070] d5: on-axis thickness of the second lens **15**;

[0071] d6: negative value of the on-axis thickness of the second lens **15**;

[0072] d7: negative value of the on-axis distance from the front side surface **141** of the quarter wave plate **14** to the rear side surface **151** of the second lens **15**;

[0073] d8: negative value of the on-axis thickness of the quarter wave plate **14**;

[0074] d9: on-axis distance of the front side surface **152** of the second lens **15** to the rear side surface **171** of the third lens **17**;

[0075] d10: on-axis thickness of the third lens **17**;

[0076] d11: on-axis distance from the front side surface **172** of the third lens **17** to the rear side surface **181** of the circular polarizer **18**;

[0077] d12: on-axis thickness of the circular polarizer **18**;

[0078] nd: refractive index of the d line (the d line is green light with a wavelength of 540 nm);

[0079] nd1: refractive index of the d line of the first lens **12**;

[0080] nd2: refractive index of the d line of the second lens **15**;

[0081] nd3: refractive index of the d line of the third lens **17**;

[0082] ng: refractive index of the d line of the circular polarizer **18**;

[0083] vd: Abbe number;

[0084] v1: Abbe number of the first lens **12**;

[0085] v2: Abbe number of the second lens **15**;

[0086] v3: Abbe number of the third lens **17**;

[0087] vg: Abbe number of the circular polarizer **18**.

[0088] Table 2 shows the data of the aspherical surface of each lens in the optical system **100** according to the first embodiment of the present disclosure.

TABLE 2

	Conic coefficient	Aspheric surface coefficients			
	k	A4	A6	A8	A10
R1	1.25E+01	−5.52E−06	2.52E−09	2.63E−12	−3.55E−14
R2	−5.42E+01	2.55E−06	2.03E−09	1.41E−12	−1.79E−16
R3	−9.90E+01	8.27E−08	−2.72E−09	−1.79E−12	−2.61E−15
R4	−1.24E+00	4.49E−07	−2.87E−11	2.37E−12	1.26E−15
R5	5.19E+01	7.41E−06	4.93E−08	2.95E−11	5.78E−14
R6	−6.03E+01	4.05E−06	5.17E−09	1.23E−12	−7.01E−14

	Conic coefficient	Aspheric surface coefficients		
	k	A12	A14	A16
R1	1.25E+01	−2.00E−16	−3.00E−19	1.82E−21
R2	−5.42E+01	−3.73E−18	−4.20E−21	−1.15E−23
R3	−9.90E+01	−3.03E−18	−1.24E−20	−1.57E−23

TABLE 2-continued

R4	−1.24E+00	−2.42E−18	−6.26E−21	−2.73E−23
R5	5.19E+01	−5.07E−17	5.40E−19	1.58E−21
R6	−6.03E+01	−2.85E−16	−5.83E−19	5.29E−21

[0089] For convenience, the aspherical surfaces of each lens surfaces use the aspherical surfaces shown in formula (1) below. However, the present disclosure is not limited to the polynomial form of the aspherical surface expressed in this formula (1).

$$z = (cr^2)/\left\{1 + \left[1(k+1)(c^2r^2)\right]^{1/2}\right\} + A4r^4 + A6r^6 + A8r^8 + A10r^{10} + A12r^{12} + A14r^{14} + A16r^{16} \tag{1}$$

[0090] k represents the conic coefficient, A4, A6, A8, A10, A12, A14 and A16 represent the aspheric surface coefficients, C represents the curvature radius at the center of the optical surface, R represent a vertical distance between the point on the aspheric curve and the optical axis, and Z represent an aspheric depth (a vertical distance between the point on the aspheric surface at a distance of r from the optical axis and the tangent plane tangent to the vertex on the optical axis of the aspheric surface).

[0091] FIG. **2** and FIG. **3** illustrate a diagram of a dot column and a diagram of a lateral color of light with wavelengths of 470 nm, 540 nm and 630 nm after passing through the optical system **100** according to the first embodiment, respectively. FIG. **4** illustrates a schematic diagram of a field curvature and a distortion of light with a wavelength of 540 nm after passing through the optical system **100** according to the first embodiment. A field curvature S in FIG. **4** is a field curvature in a sagittal direction, and T is a field curvature in a meridian direction.

[0092] In this embodiment, an entrance pupil diameter ENPD of the optical system **100** is 4.00 mm, an image height IH of 1.0H is 11.500 mm, and a field of view FOV in a diagonal direction is 89.94°. Thus, the optical system **100** meets the design requirements of miniaturization and light-weight, and the on-axis chromatic aberration and the off-axis aberration are fully corrected, thereby achieving excellent optical characteristics.

Second Embodiment

[0093] The second embodiment is substantially the same as the first embodiment, the symbols have the same meaning as the first embodiment, and only the differences are listed below.

[0094] An optical system **200** according to a second embodiment of the present disclosure is shown in FIG. **6** and FIG. **10**.

[0095] As shown in FIG. **10**, in the optical system **200** according to the second embodiment, the composite film including reflective polarizing coating **13** and the quarter-wave plate **14** is arranged on the rear side surface **151** of the second lens **15**.

[0096] Table 3 and table **4** show the design data of the optical system **200** according to the second embodiment of the present disclosure.

[0097] Since the arrangement position of the composite film in the second embodiment is different from the arrangement position of the composite film in the first embodiment, unlike the first embodiment, d4 in Table 3 represents: the on-axis distance from the front side surface **122** of the first lens **12** to the rear side surface **131** of the reflective polarizing coating **13**.

TABLE 3

	R		d		nd		vd
OBJECT	∞		−1342.8				
Aperture	∞	d0	13.800				
R1	−50.139	d1	1.671	n1	1.658	v1	21.00
R2	−32.608	d4	0.249				
R2	−49.194	d2 + d3	0.300		1.492		57.44
R2	−49.194	d5	5.647	n2	1.544	v2	56.11
R3	−30.903	d6	−5.647				
R4	−49.194	d8	−0.127		1.492		57.44
R3	−49.194	d3	0.127				
R2	−49.194	d5	5.647	n2	1.544	v2	56.11
R2	−30.903	d9	0.350				
R2	−216.350	d10	8.009	n3	1.658	v3	21.00
R3	−79.264	d11	1.200				
Circular polarizer		d12	0.550	ng	1.517	vg	64.21
Image surface							

[0098] Table 4 shows the data of the aspherical surface of each lens in the optical system **200** according to the second embodiment of the present disclosure.

TABLE 4

	Conic coefficient	Aspheric surface coefficient			
	k	A4	A6	A8	A10
R1	−2.97E+01	−6.25E−06	−1.78E−08	7.50E−11	−7.25E−14
R2	−1.02E+01	2.91E−06	2.62E−08	−2.93E−11	1.09E−13
R3	−4.32E+00	7.16E−07	3.84E−09	1.20E−11	4.76E−14
R4	1.67E−01	1.22E−06	5.60E−09	7.78E−12	1.44E−14
R5	8.29E+01	3.19E−05	−5.12E−08	1.54E−10	2.63E−13
R6	−7.12E−01	−1.38E−05	5.60E−08	1.37E−10	−4.13E−13

TABLE 4-continued

	Conic coefficient	Aspheric surface coefficients		
	k	A12	A14	A16
R1	−2.97E+01	−3.48E−16	−1.01E−17	3.28E−20
R2	−1.02E+01	−1.09E−16	−1.26E−18	1.23E−21
R3	−4.32E+00	−4.45E−17	2.02E−20	−1.70E−22
R4	1.67E−01	3.99E−17	2.03E−19	−2.05E−22
R5	8.29E+01	7.01E−16	−2.12E−20	−1.38E−20
R6	−7.12E−01	−3.02E−15	6.18E−18	2.13E−21

[0099] FIG. **7** and FIG. **8** illustrate a diagram of a dot column and a diagram of a lateral color of light with wavelengths of 470 nm, 540 nm and 630 nm after passing through the optical system **200** according to the second embodiment, respectively. FIG. **9** is a schematic diagram of a field curvature and a distortion of light with a wavelength of 540 nm after passing through the optical system **200** according to the second embodiment. A field curvature S in FIG. **9** is a field curvature in a sagittal direction, and T is a field curvature in a meridian direction.

[0100] In this embodiment, an entrance pupil diameter ENPD of the optical system **100** is 4.00 mm, an image height IH of 1.0H is 11.200 mm, and a field of view FOV in a diagonal direction is 94.95°. Thus, the optical system **200** meets the design requirements of miniaturization and lightweight, and the on-axis aberration and the off-axis aberration are fully corrected, thereby achieving excellent optical characteristics.

TABLE 5

Parameters and Conditional Equations	First Embodiment	Second Embodiment
f2/f	5.19	8.93
(R1 + R2)/(R1 − R2)	1.38	4.72
R5/R6	0.94	2.73
SDmax	23.00	22.00
eyebox	12.00	12.00
TL	18.307	17.977
TTL	34.707	31.777
IH	11.500	11.200
FOV	89.94	94.95

[0101] It can be understood by those skilled in the art that the above embodiments are specific embodiments for realizing the present disclosure. However, in practical application, various changes can be made in form and detail without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An optical system, comprising in sequence from a rear side to a front side:

an aperture, located on the rear side of the optical system;

a first lens;

a second lens;

a third lens;

a circular polarizer; and

an image surface;

wherein the circular polarizer is attached to a rear side surface of the image surface, a front side surface of the first lens or a rear side surface of the second lens is provided with a composite film including a reflective polarizing coating and a quarter-wave plate, and the reflective polarizing coating is applied on a rear side of the quarter-wave plate;

wherein the optical system satisfies the following conditions:

$$5.00 \le f2/f \le 9.00;$$

$$1.30 \le (R1 + R2)/(R1R2) \le 4.80;$$

$$0.90 \le R5/R6 \le 2.80; \text{ and}$$

$$SD\mathrm{max} \le 23.00 \text{ mm}.$$

wherein f represents a focal length of the optical system;

f2 represents a focal length of the second lens;

R1 represents a central curvature radius of a rear side surface of the first lens;

R2 represents a central curvature radius of the front side surface of the first lens;

R5 represents a central curvature radius of a rear side surface of the third lens;

R6 represents a central curvature radius of a front side surface of the third lens; and

SDmax represents a maximum effective radius of lenses of the optical system.

2. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

eyebox≥12 mm;

wherein eyebox represents a size of an eye box of the optical system.

3. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

eyerelief≤16.50 mm;

wherein eyerelief represents an on-axis distance from a human eye to the rear side surface of the first lens.

4. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

$TL \le 18.40$ mm;

wherein TL represents an on-axis distance from the rear side surface of the first lens to the image surface.

5. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

$TTL \le 34.80$ mm;

wherein TTL represents a total track length TTL of the optical system.

6. The optical system of claim **1**, wherein the rear side surface and the front side surface of the first lens, the rear side surface and a front side surface of the second lens, and the rear side surface and the front side surface of the third lens are aspherical.

7. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

$$85.00° \leq FOV \leq 95.00°;$$

wherein FOV represents a field of view of the optical system.

8. The optical system of claim **1**, wherein a front side surface of the second lens is coated with a semi-transparent and semi-reflective film.

9. The optical system of claim **8**, wherein each of a transmittance and a reflectivity of the semi-transparent and semi-reflective film ranges from 40% to 60%.

10. The optical system of claim **1**, wherein a transmittance of the reflective polarizing coating is greater than or equal to 95%.

11. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

$$MIST \leq 35.00\%;$$

wherein MIST represents a distortion of the optical system.

12. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

$$LC \leq 100\ \mu m;$$

wherein LC represents a lateral color of the optical system.

13. The optical system of claim **1**, wherein the optical system further satisfies the following condition:

$$TTL/f \leq 2.10;$$

wherein TTL represents a total track length of the optical system.

14. The optical system of claim **1**, wherein the image surface is a display with a size ranging from 1.0 inch to 1.3 inches.

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